
QF604 Econometrics of Financial Markets

Multivariate regression 1 & 2 - Lectures 4 & 5

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Outline

- Omitted variable bias (OVB)
- Multivariate regression: Assumptions and Asymptotic properties
- Goodness of fit (R^2 and adjusted- R^2)
- An empirical finance example
- Linear algebra for multivariate regression - inference

Omitted variable bias (OVB)

- The methodology we've covered so far has (at least) one big limitation: there's only one variable "explaining" Y
- Consider the stock price regression:

```
. reg stock price book value, robust
```

Regression with robust standard errors

					Number of obs =	420
					F(1, 418) =	19.26
					Prob > F	= 0.0000
					R-squared	= 0.0512
					Root MSE	= 18.561

		Coef.	Std. Err.	Robust t	P> t	[95% Conf. Interval]	
stock price							
book value		-2.279808	0.5194892	-4.39	0.000	-3.300945	-1.258671
_cons		698.933	10.36436	67.44	0.000	678.5602	719.3057

- What if *book value* is picking up something else besides itself?

Omitted variable bias, cont'd

- In other words, what if something else is driving the price of the stock? For example, earnings-per-share (eps), leverage,...
- Why do we care? We'd like to establish a *causal* effect. And we don't want *book value* getting credit (or blame) for the effect of something else!
- Even worse, what if *book value* is significant only because *other variables* are correlated with both *book value* and *price*?
- Both problems are examples of **omitted variable bias**
- Definition: If a regressor is correlated with a variable that has been omitted from the analysis but that determines (in part) the dependent variable, then the OLS estimator will have omitted variable bias

Omitted variable bias, cont'd

- Omitted variable bias (OVB) occurs when two conditions hold:
 1. The OV is correlated with the included regressor (OVB 1)
 2. The OV is a determinant of the dependent variable (OVB 2)
- Classical example:
 - Price of stock and GDP

$$Price = \beta_0 + \beta_1 book_value + u$$

Omitting *eps* might cause you to overestimate the importance of *book_value* (*eps* bias). Can you see why?

Omitted variable bias, cont'd

- Let's see what happens when we omit a relevant variable from our analysis. Suppose the *true* model is:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i \quad (1)$$

with $E[u_i | X_{1i}, X_{2i}] = 0$. So β_1 is the true slope of X_1

- Note that, by the LIE, $E[u_i | X_{1i}] = E[E[u_i | X_{1i}, X_{2i}] | X_{1i}] = 0$
- Now suppose we (incorrectly) assume

$$Y_i = \alpha_0 + \alpha_1 X_{1i} + v_i \quad (2)$$

where $E[v_i | X_{1i}] = 0$

- Given the true model of Y_i , what is v_i here? $v_i = \beta_2 X_{2,i} + u_i$
- If we incorrectly run the univariate OLS, which OLS assumption is violated? The correct specification assumption is violated. $E(v_i | X_{1,i}) = \beta_2 E(X_{2,i} | X_{1,i}) \neq 0$.

Omitted variable bias, cont'd

- However, we can still estimate α_1 , the slope of X_1 , using equation (2). The OLS estimator $\hat{\alpha}_1$ (which minimizes the sum of squared prediction errors) is:

$$\hat{\alpha}_1 = \frac{s_{X_1Y}}{s_{X_1}^2} = \frac{\sum (X_{1i} - \bar{X}_1)(Y_i - \bar{Y})}{\sum (X_{1i} - \bar{X})^2} = \beta_1 + \beta_2 \frac{s_{X_1X_2}}{s_{X_1}^2} + \frac{s_{X_1u}}{s_{X_1}^2}$$

- How does $\hat{\alpha}_1$ relate to β_1 ? In particular, is $E(\hat{\alpha}_1) = \beta_1$, i.e. is $\hat{\alpha}_1$ unbiased? Does $\hat{\alpha}_1 \xrightarrow{p} \beta_1$, i.e. is $\hat{\alpha}_1$ consistent?

Consistency

- When you have OVB, OLS is inconsistent - this is a more serious issue!
Can you derive the probability limit of $\hat{\alpha}_1$?

- From the previous derivation, we see that

$$\hat{\alpha}_1 = \beta_1 + \beta_2 \frac{s_{X_1 X_2}}{s_{X_1}^2} + \frac{s_{X_1 u}}{s_{X_1}^2}$$

- We know $s_{X_1 X_2} \xrightarrow{p} \sigma_{X_1 X_2}$, $s_{X_1}^2 \xrightarrow{p} \sigma_{X_1}^2$, and $s_{X_1 u} \xrightarrow{p} \sigma_{X_1 u} = 0$

- Therefore:

$$\hat{\alpha}_1 \xrightarrow{p} \beta_1 + \beta_2 \frac{\sigma_{X_1 X_2}}{\sigma_{X_1}^2} \neq \beta_1 \text{ unless } \sigma_{X_1 X_2} = 0 \text{ (or } \beta_2 = 0 \text{)}.$$

Omitted variable bias, cont'd

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

- The omitted variable is correlated with the included regressor (OVB 1): $\sigma_{X_1 X_2} \neq 0$.
- The omitted variable is a determinant of the dependent variable (OVB 2): $\beta_2 \neq 0$.

Direction of the bias

- The correlation between the regressor and the omitted variable (more generally, the error term) determines the direction (sign) of the bias:

$$\hat{\alpha}_1 \xrightarrow{p} \beta_1 + \beta_2 \frac{\sigma_{X_1 X_2}}{\sigma_{X_1}^2}$$

- Since $\sigma_{X_1}^2$ is positive, the signs of β_2 and $\sigma_{X_1 X_2}$ determine the direction (sign) of the bias term $(\beta_2 \frac{\sigma_{X_1 X_2}}{\sigma_{X_1}^2})$.

Some things to remember about OVB

- Omitted regressors are absorbed by the error term, e.g. $v = \beta_2 X_2 + u$
- OLS Assumption 1 (correct specification) does not hold, $E(v|X_1) \neq 0$
- The OLS estimator is biased and inconsistent, OVB is thus a problem whether the sample size is large or small
- The magnitude of the bias depends on the correlation between the regressor and the omitted variable or, more generally, the error term
 - No problem if X_1 and X_2 are uncorrelated (X_1 can't “absorb” the effect of X_2 on Y). Same if X_2 is not correlated with Y
- The correlation between the regressor and the omitted variable (more generally, the error term) determines the direction (sign) of the bias
 - Why? Since σ_u & σ_X are both positive, ρ_{Xu} determines the sign

A classical example from Finance

- Will higher *profitability ratio* increase the returns of a stock?
- *Stock returns* are in $\times 100\%$

```
. reg stock returns profitability ratio, robust
```

Regression with robust standard errors

Number of obs = 935
F(1, 933) = 95.65
Prob > F = 0.0000
R-squared = 0.1070
Root MSE = 382.32

	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
profitability ratio	60.21428	6.156956	9.78	0.000	48.1312 72.29737
_cons	146.9524	80.26953	1.83	0.067	-10.57731 304.4822

- So an increase in *profitability ratio* increases the returns by 0.0602%

- What do you think is missing here?
- How about leverage? Can we hope to measure it? We could use a stock's $\frac{Debt}{Asset}$ ratio as a proxy (since we can get data on this)

```
. reg price volume, robust
```

Regression with robust standard errors

Number of obs =	420
F(1, 418) =	19.26
Prob > F =	0.0000
R-squared =	0.0512
Root MSE =	18.581

	price		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]
	volume		-2.279808	.5194892	-4.39	0.000	-3.300945 -1.258671
	_cons		698.933	10.36436	67.44	0.000	678.5602 719.3057

sure enough, $Debt/Asset$ and *profitability ratio* are positively correlated.

- What do you think will happen to the coefficient of *profitability ratio* if

we control for *Debt/Asset*?

$$\hat{\alpha}_1 \xrightarrow{p} \beta_1 + \beta_2 \frac{\sigma_{X_1 X_2}}{\sigma_{X_1}^2}$$

```
. reg stock_returns Debt_Asset Profitability_Ratio, robust
```

N = 935 F(2,932) = 64.47 Prob > F = 0.0000
R-squared = 0.1339 Root MSE = 376.73

stock_returns	Coef.	Std.Err.	t	P> t	[95%CI]
Debt_Asset	42.0576	6.8101	6.18	0.000	[28.6928, 55.4225]
Profit_Ratio	5.1380	0.9266	5.54	0.000	[3.3194, 6.9565]
_cons	-128.8899	93.0940	-1.38	0.167	[-311.5879, 53.8082]

- Now we have estimated the returns to *profitability ratio*, *controlling for leverage*. The expected impact of profitability ratio on returns is now smaller. Before it was getting credit both for the profitability **and** for differences in leverage

-
- So how do we do this in general, and how does it solve the omitted variable bias problem?

*Next:
Multivariate regression:
assumptions and asymptotic
properties*

Multiple regression

- The simplest solution to the omitted variable bias problem is to add other relevant variables to the regression. However, this solution is most often imperfect...

- Example:

$$Stock_returns = \beta_0 + \beta_1 profitability_ratio + \beta_2 leverage + u$$

- How do we add?
- Fortunately, adding additional regressors is really easy... as long as we can obtain them from the data!

Estimation of the multiple regression model

- In the multiple regression model, we assume that the population regression **line** (the relationship that holds between Y and all the X 's on average) is given by

$$E(Y_i|X_{1i}, \dots, X_{ki}) = \beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki}$$

- As in the univariate case, β_0 is the intercept and the expected value of Y_i when all the regressors (X_k 's) are zero; β_k is the slope coefficient of X_k
- β_1 (the slope coefficient of X_1) is the effect on Y (the expected change in Y) of a one unit change in X_1 , **holding all other variables constant** (or “controlling for all other variables”)
- β_1 may also be described as the *partial effect* on Y of X_1

Estimation of the multiple regression model, ctd.

- The population regression **model** is then given by

$$Y_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki} + u_i, \text{ where}$$

$$u_i \equiv Y_i - (\beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki}) = Y_i - E(Y_i | X_{1i}, \dots, X_{ki})$$

- The true meaning of assuming the population regression line is that, under this assumption, $E(u_i | X_{1i}, \dots, X_{ki}) = 0$
- As in the univariate case, the OLS residuals are given by $\hat{u}_i = Y_i - \hat{Y}_i$, where $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \dots + \hat{\beta}_k X_{ki}$
- OLS minimizes the sum of the squared errors $(\sum \hat{u}_i^2)$, yielding closed form formulas for the estimators $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$
- Since the formulas involve matrix algebra, we will not derive them here, but the intuition is the same as in the univariate case

OLS assumptions

- As in the univariate case, we must make some assumptions in

$$Y_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki} + u_i$$

OLS Assumption 1 Correct specification

$$E(u_i | X_{1i}, \dots, X_{ki}) = 0$$

OLS Assumption 2 Simple random sample

$(Y_i, X_{1i}, \dots, X_{ki})$ are *iid* draws from their joint distribution

OLS Assumption 3 No extreme outliers

$X_{1i}, \dots, X_{ki}, u_i$ have non-zero and finite fourth moments

OLS Assumption 4 No perfect collinearity

Regressors are not linear combinations of each other

OLS Assumption 4: no perfect collinearity

- Assumption 4 is new! Why is it important?
- Suppose $X_{2i} = a + bX_{1i}$ (where a and b are known constants)
$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i = (\beta_0 + \beta_2 a) + (\beta_1 + \beta_2 b) X_{1i} + u_i$$
- This is equivalent to the univariate regression, which we were able to estimate because the FOCs gave us two equations and two unknowns
- Now there are three unknowns $(\beta_0, \beta_1, \beta_2)$, but still only two equations, so we can't *identify* the parameters
- An alternative way to think about the problem: we can find multiple vectors of parameters $(\beta_0, \beta_1, \beta_2)$ which will give the same regression line! All of them will be consistent with the data

OLS Assumption 4: no perfect collinearity, cont'd

- Examples of perfect collinearity
 - Fraction of English learners and % of English speakers, dummies for males and females...
 - Height in centimeter., height in inches...
- In practice software like Python will give you an error message. Eviews and Stata will delete some of the collinear variables. **However, it is better to drop them beforehand.**

Imperfect multicollinearity

- OLS A4 rules out perfect collinearity. But what if two or more of the regressors are highly (but not perfectly) collinear?
- In other words, what if there's a linear function of the regressors that is highly correlated with another regressor?
 - Example: Hedonic Regression - Regressing the price of a good on its characteristics (e.g. cars, diamonds, houses). How different are horsepower and acceleration?
- This situation, called imperfect multicollinearity, does not pose any problem for the *theory* of the OLS estimators
- In fact, a purpose of OLS is to sort out the independent effects of the various regressors when they are potentially correlated

Imperfect multicollinearity, cont'd

- **BUT** with a high degree of imperfect multicollinearity among the regressors, $\sigma_{\hat{\beta}_j}$ will tend to be high (at least among the correlated regressors)
 - This can make it difficult to obtain precise estimates of the separate effects (especially in small samples)
- Intuition: recall that a coefficient is the partial effect of one regressor, holding the other ones constant
 - If the regressors tend to move together, this effect will be hard to estimate precisely

- This can be illustrated using simulated data

```
. clear
. set obs 1000
obs was 0, now 1000
. gen x=uniform()
. gen x2=x+0.001*invnorm(uniform())
. gen y=4+3*x+invnorm(uniform())
. reg y x x2
```

Source	SS	df	MS	Number of obs = 1000		
Model	795.758978	2	397.879489	F(2, 997) = 382.40		
Residual	1037.3519	997	1.04047332	Prob > F = 0.0000		
				R-squared = 0.4341		
				Adj R-squared = 0.4330		
Total	1833.11087	999	1.83494582	Root MSE = 1.02		

y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
x	11.21924	33.24992	0.34	0.736	-54.02861	76.46708
x2	-8.113489	33.24836	-0.24	0.807	-73.35828	57.1313
_cons	3.94279	.0645122	61.12	0.000	3.816194	4.069385

```
.
```

More on (imperfect) multicollinearity

- General principle:
 - Multicollinearity exists in almost all regressions
 - Virtually impossible in the real world to find a set of independent variables that are totally uncorrelated with each other
 - Our purpose is to learn to determine how much multicollinearity exists by using two general indicators or diagnostic tools

Indicator 1: t-test v.s. R^2

- Look for a 'high' R^2 , but 'few' significant t ratios
 - Common 'rule of thumb': Can't reject the null hypotheses that coefficients are individually equal to zero (t tests), but can reject the null hypothesis that they are simultaneously equal to zero (F test, to be discussed in detail later)
 - This is not an exact test. What do we mean by 'few' significant t ratios, and a 'high' R^2 ? Too imprecise. Also depends on other factors like the sample size

Indicator 2: correlation matrix of regressors

- Look for high pair-wise correlation coefficients. Look at the correlation matrix for the regressors.
 - How high is high? As a rule of thumb, we can use 0.8. If the sample correlation exceeds 0.8 in absolute value, we should be concerned about multicollinearity
 - Multicollinearity refers to a linear relationship among all or some of the regressors. Any pair of independent variables may not be highly correlated, but one variable may be a linear function of a number of others. In a 2-variate regression, multicollinearity is the correlation between the 2 explanatory variables
 - This is a “sufficient, but not a necessary condition for multicollinearity.” In other words, if you’ve got a high pairwise correlation, you’ve got problems. However, it isn’t conclusive evidence of an absence of multicollinearity

Multicollinearity: an example

- Example: estimating daily stock returns using two highly related market factors (2010–2024):

$$R_{i,t} = \beta_0 + \beta_1 R_{m,t} + \beta_2 R_{s,t} + \varepsilon_t$$

where: $R_{i,t}$ = daily return on stock i ; $R_{m,t}$ = daily return on a broad market index (e.g., S&P 500); $R_{s,t}$ = daily return on a very similar index (e.g., a large-cap ETF tracking the same market)

- Suppose the regression output is:

$$\hat{R}_{i,t} = 0.0002 + \underset{(0.15)}{1.10} R_{m,t} - \underset{(0.14)}{0.95} R_{s,t}$$
$$n = 250, \quad R^2 = 0.92.$$

- Multicollinearity is a problem here. Both t-ratios are small (despite a very high R^2).

-
- It is easy to confirm multicollinearity in this case. The correlation between $R_{m,t}$ and $R_{s,t}$ is extremely high (≈ 0.92) because the two series track nearly the same underlying risk.
 - Standard errors blow up. We can't separately identify the exposure to the market factor versus the "almost-the-same" index factor.
 - One possible solution: Panel Data, which we will learn soon

A priori information (Covered Interest Parity)

- If we know a relationship between slope coefficients from financial theory, we can impose it to reduce multicollinearity.
- Example: Covered Interest Parity (CIP) links the forward premium to the interest rate differential.
 - Let $s_t = \log S_t$ be the log spot exchange rate (domestic price of foreign currency), and $f_t = \log F_t$ be the log forward rate for the same maturity. Let $i_{d,t}$ and $i_{f,t}$ be domestic and foreign interest rates (same maturity).
 - CIP (approximate log form) implies:

$$f_t - s_t \approx i_{d,t} - i_{f,t}.$$

-
- If we try to estimate the relationship using *both* interest rates:

$$f_t - s_t = \beta_0 + \beta_1 i_{d,t} + \beta_2 i_{f,t} + \varepsilon_t,$$

we may face multicollinearity because $i_{d,t}$ and $i_{f,t}$ often move together and can be highly correlated.

- Theory gives a restriction on coefficients:

$$\beta_1 + \beta_2 = 0 \quad (\text{i.e., } \beta_2 = -\beta_1),$$

and under CIP exactly we would also have $\beta_0 = 0$ and $\beta_1 = 1$.

Imposing the restriction

- Regression:

$$f_t - s_t = \beta_0 + \beta_1 i_{d,t} + \beta_2 i_{f,t} + \varepsilon_t$$

- Impose the theory restriction $\beta_2 = -\beta_1$:

$$f_t - s_t = \beta_0 + \beta_1 i_{d,t} - \beta_1 i_{f,t} + \varepsilon_t$$

- Rearranging:

$$f_t - s_t = \beta_0 + \beta_1 (i_{d,t} - i_{f,t}) + \varepsilon_t$$

- Now you estimate only $\hat{\beta}_1$ using the interest differential, and set $\hat{\beta}_2 = -\hat{\beta}_1$ by construction.

Get more data

- Extend the data set: Multicollinearity is a “sample phenomenon”
- Change Nature or Source of Data.
 - We can switch from cross-sectional analysis to panel data analysis
 - The use of panel data potentially reduces the multicollinearity in the total sample
 - E.g., a random sample of households over a number of years
 - E.g., the stocks in S&P 500 over a period

Do nothing (A remedy!)

- Imperfect multicollinearity is not a problem if the objective of the analysis is forecasting. It doesn't affect the overall explanatory power of the regression (i.e., R^2)
- It is a problem if the objective is to test the significance of individual coefficients because of the inflated variances/standard errors

OLS Assumption 5?

- The concepts of **heteroskedasticity** and **homoskedasticity** also carry over to the multiple regression model
- If we want to assume homoskedasticity (in general we won't), we would add:

OLS Assumption 5 Homoskedasticity

$$\text{var}(u_i | X_{1i}, \dots, X_{ki}) = \sigma_u^2$$

Properties of multivariate OLS

- Assumptions A1-A4 are sufficient to prove that OLS yields an **unbiased** and **consistent** estimator of the intercept and all the slopes
- They're also sufficient to prove that the $\hat{\beta}$'s are **asymptotically normal**
- In this case, this means that not only for each coefficient

$$\hat{\beta}_j \stackrel{a}{\sim} N\left(\beta_j, \hat{\sigma}_{\hat{\beta}_j}^2\right)$$

but all the estimated coefficients are **jointly normal** (asymptotically)

- Adding OLS A5 makes OLS BLUE

Next: Goodness of fit

Goodness of fit

```
. reg price volume, robust
```

Regression with robust standard errors

Number of obs = 420
 F(1, 418) = 19.26
 Prob > F = 0.0000
 R-squared = 0.0512
 Root MSE = 18.581

		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
price							
volume		-2.279808	.5194892	-4.39	0.000	-3.300945	-1.258671
_cons		698.933	10.36436	67.44	0.000	678.5602	719.3057

```
. reg price industry proxy volume, robust
```

Regression with robust standard errors

Number of obs = 420
 F(2, 417) = 223.82
 Prob > F = 0.0000
 R-squared = 0.4264
 Root MSE = 14.464

		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
price							
industry proxy		-1.101296	.4328472	-2.54	0.011	-1.95213	-.2504616
volume		-.6497768	.0310318	-20.94	0.000	-.710775	-.5887786
_cons		686.0322	8.728224	78.60	0.000	668.8754	703.189

Goodness of fit, cont'd

- You may have noticed that the R^2 went way up when we added *industry_proxy* to the regression
- If we use our reasoning from the univariate case, $R^2 = 42.6\%$ means that we are explaining about 42.6% of the variation in student test score with our regression, up from about 5.1% with only *volume*
- So it seems that we are doing a much better job of explaining test scores now. But is this logic valid?

Goodness of fit, cont'd

- The answer is yes, but we need to revisit the notion of *goodness of fit* to convince ourselves of why
- To assess goodness of fit, we ask how well our OLS estimates fit the data?
- There are 3 ways to answer this question
 - The standard error of the regression (SER), omitted
 - R^2
 - *Adjusted- R^2*

R-squared

- As before, the regression R^2 is the fraction of the sample variance of Y_i explained by the regressors,

$$R^2 = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2} = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS} = 1 - \frac{\sum (Y_i - \hat{Y}_i)^2}{\sum (Y_i - \bar{Y})^2}$$

- However, in multiple regression, the R^2 increases whenever a regressor is added (unless it's perfectly multicollinear with the original regressors)
- Recall that OLS is picking the $\hat{\beta}$'s to minimize the SSR. When the k^{th} regressor is added, it can always set $\hat{\beta}_k = 0$, making it no worse than you did with $k - 1$ regressors. It's rare to get an estimate being exactly zero. In general, SSR will decrease when a regressor is added
- So an unscrupulous econometrician might be tempted to keep adding regressors to inflate the R^2

Adjusted R-squared

- One way to adjust for this is to deflate the R^2 by some factor, which is what the *adjusted- R^2* or $\bar{R}^2$ does,

$$\bar{R}^2 = 1 - \frac{n-1}{n-k-1} \frac{SSR}{TSS} = 1 - \frac{s_{\hat{u}}^2}{s_Y^2}$$

Notice that:

1. $\bar{R}^2$ is always less than R^2 : $\frac{n-1}{n-k-1} > 1 \implies \bar{R}^2 < R^2$
2. Adding a regressor has two effects on $\bar{R}^2$: 1.) SSR falls, but, 2.) $\frac{n-1}{n-k-1}$ increases. So the total effect on $\bar{R}^2$ depends on which effect is bigger
3. $\bar{R}^2$ can be negative (although rare in practice)

Adjusted R-squared, cont'd

- To see how $\bar{R}^2$ works, let's see what happens when we add irrelevant regressors to the Test Scores regression
- We create three independent $N(0, 1)$ variables (junk1-junk3) and then add them to the regression of *price* on *volume* and *industry_proxy*
- Since these new variables are randomly generated, they shouldn't help explain any of the variation in prices of the stock.
- Let's see what happens in practice

Source	SS	df	MS	Number of obs = 420		
Model	64864.3011	2	32432.1506	F(2, 417) = 155.01		
Residual	87245.2925	417	209.221325	Prob > F = 0.0000		
				R-squared = 0.4264		
				Adj R-squared = 0.4237		
Total	152109.594	419	363.030056	Root MSE = 14.464		
price	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
industry proxy	-1.101296	.3802783	-2.90	0.004	-1.848797	-.3537946
volume	-.6497768	.0393425	-16.52	0.000	-.7271112	-.5724423
_cons	686.0322	7.411312	92.57	0.000	671.4641	700.6004
Source	SS	df	MS	Number of obs = 420		
Model	64906.6894	3	21635.5631	F(3, 416) = 103.21		
Residual	87202.9042	416	209.622366	Prob > F = 0.0000		
				R-squared = 0.4267		
				Adj R-squared = 0.4226		
Total	152109.594	419	363.030056	Root MSE = 14.478		
price	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
industry proxy	-1.094981	.3809016	-2.87	0.004	-1.843712	-.3462489
volume	-.6502859	.0393965	-16.51	0.000	-.7277269	-.5728449
junk1	-.335101	.7451975	-0.45	0.653	-1.799923	1.129721
_cons	685.9061	7.423711	92.39	0.000	671.3135	700.4988
Source	SS	df	MS	Number of obs = 420		
Model	65010.6439	4	16252.661	F(4, 415) = 77.44		
Residual	87098.9497	415	209.876987	Prob > F = 0.0000		
				R-squared = 0.4274		
				Adj R-squared = 0.4219		
Total	152109.594	419	363.030056	Root MSE = 14.487		
price	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
industry proxy	-1.097402	.3811464	-2.88	0.004	-1.846624	-.3481797
volume	-.651411	.0394128	-16.51	0.000	-.7288833	-.5738587
junk1	-.3239416	.7458185	-0.43	0.664	-1.789995	1.142111
junk2	.4838163	.6874502	0.70	0.482	-.8675022	1.835135
_cons	685.0658	7.423702	92.34	0.000	671.3832	700.5084
Source	SS	df	MS	Number of obs = 420		
Model	65335.3015	5	13067.0603	F(5, 414) = 62.34		
Residual	85774.2921	414	209.599739	Prob > F = 0.0000		
				R-squared = 0.4295		
				Adj R-squared = 0.4226		
Total	152109.594	419	363.030056	Root MSE = 14.478		
price	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
industry proxy	-1.105399	.3809508	-2.90	0.004	-1.854238	-.3565603
volume	-.6511835	.0394272	-16.52	0.000	-.7286859	-.5736811
junk1	-.3242784	.7453236	-0.44	0.664	-1.788373	1.140816
junk2	.5627173	.6899149	0.82	0.415	-.7934557	1.91889
junk3	.851649	.6879098	1.24	0.214	-.4966934	2.2008
_cons	686.1817	7.425821	92.40	0.000	671.5847	700.7787

Some caveats: as before

1. An increase in R^2 or $\bar{R}^2$ does not mean that the added variable is statistically significant
 - R^2 always increases when we add regressors. $\bar{R}^2$ might not, but even if it does, it doesn't mean that the added variable is statistically significant. You need to run a hypothesis testing to establish this.
2. A high R^2 or $\bar{R}^2$ does not mean that the regressors are a true cause of the dependent variable
3. A high R^2 or $\bar{R}^2$ does not mean that there is no omitted variable bias

-
4. A high R^2 or $\bar{R}^2$ does not necessarily mean that you have the most appropriate set of regressors, nor does a low R^2 or $\bar{R}^2$ necessarily mean that you have a “bad” set of regressors

Next: An example

An example: News sentiment and financial markets

- Research question: Do firms with more positive news sentiment earn higher stock returns (or face lower financing costs)?
- Example paper: Paul C. Tetlock (2007) “Giving Content to Investor Sentiment: The Role of Media in the Stock Market,” *Journal of Finance*, vol.62(3), pp.1139–1168
- Article available on JSTOR

Does sentiment matter in financial markets?

- Is media tone / investor sentiment systematically related to stock returns, volatility, or trading volume?
- If we learn that returns are related to sentiment, do we have evidence of mispricing (inefficiency), or does sentiment proxy for risk or information?
- Common-sense definition of mispricing:
Price moves driven by non-fundamental factors rather than changes in intrinsic value

Fundamentals and asset prices

- What does finance theory say?
 - In efficient markets, prices reflect expected discounted cash flows and risk (discount rates)
 - Expected returns compensate investors for bearing systematic risk
- If sentiment does not contain information about fundamentals or risk, and markets are informationally efficient, then we would expect no predictable return premium associated with sentiment (after controlling for risk exposures)

Data

- Firm-level stock returns and accounting data (e.g., CRSP/Compustat; local exchange data)
- News / text data (e.g., newspaper archives, Dow Jones, RavenPack, Refinitiv News Analytics)
- Trading data (e.g., volume, bid-ask spreads, volatility measures)
- What variables do these datasets have?
 - Measure of sentiment / tone (text-based score, negative/positive word counts)

-
- Price and return variables (daily/weekly returns, abnormal returns, market-adjusted returns)
 - Firm characteristics (size, book-to-market, leverage, profitability)
 - Risk controls (market beta, factor loadings; Fama-French/Carhart factors)

Econometric model (Tetlock, 2007)

- Predict daily market returns using lagged news pessimism (distributed-lag model):

$$r_t = \alpha + \sum_{k=1}^5 \gamma_k BdNws_{t-k} + \varepsilon_t,$$

where r_t is the *daily Dow Jones return* (in basis points) and $BdNws$ is a standardized measure of negative sentiment from news text.

- We want to check whether sentiment predicts returns: test

$$H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_5 = 0.$$

Regression results (Tetlock, 2007)

- Sample: $n = 3709$ daily observations, 1984-01-01 to 1999-09-17.

Table 1: Key return-predictability estimates using negative sentiment (Tetlock, 2007, Table II)

	Pessimism factor	Negative words	Weak words
$\hat{\gamma}_1$ (lag 1)	−8.1	−4.4	−6.0
Sum of lags 2–5	6.8	9.5	10.7
Joint test $\chi^2(5)$	20.0	20.8	26.5
p -value	0.001	0.001	0.000
Reversal test $\chi^2(1)$	4.05	8.35	10.1
p -value	0.044	0.004	0.002

Conclusion of the paper

- News sentiment matters in financial markets
- Interpretations of the estimates (Tetlock, 2007)
 - Negative news predicts lower next-day returns
 - * Using the *Pessimism factor*, a 1 s.d. increase in pessimism predicts about -8.1 basis points lower return the next day
 - * Using *Negative* (or *Weak*) word content, the next-day effect is also negative (e.g., around -4.4 to -6.0 basis points)
 - Partial reversal over the next few days
 - * The sum of lags 2–5 is positive (e.g., about $+6.8$ to $+10.7$ basis points), consistent with short-run price pressure followed by reversal

*Next: Algebra of Multivariate
Regression*

Matrix Notation

- Define the K and $K + 1$ -dimensional (column) vector X_i and β respectively as

$$X_i = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{iK} \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{K+1} \end{pmatrix}$$

- Define the $n \times (K + 1)$ matrix X as

$$X = \begin{pmatrix} 1 & X'_1 \\ 1 & X'_2 \\ \vdots & \\ 1 & X'_n \end{pmatrix}$$

Matrix Notation, cont'd

- Recall that our multivariate regression model is

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_K x_{iK} + \epsilon_i$$

- Define the n -dimensional (column) vector of error ϵ and dependent variable Y as

$$\epsilon = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}, \quad Y = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}$$

- We can write it compactly in matrix form as

$$Y = X\beta + \epsilon$$

Obtaining the OLS expression

- Recall that we want to minimize $\sum_{i \in [n]} \hat{u}_i^2 \equiv \sum_{i \in [n]} (Y_i - \hat{Y}_i)^2$
- Our predicted value is $\hat{Y}_i := \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_K x_{iK}$
- We can re-express

$$\sum_{i \in [n]} \hat{u}_i^2 \equiv \sum_{i \in [n]} (Y_i - X_i' \hat{\beta})^2 = (Y - X \hat{\beta})' (Y - X \hat{\beta}),$$

$$\text{where } \hat{\beta} = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_{K+1} \end{pmatrix}$$

- How to solve this?

Obtaining the OLS expression, cont'd

- Goal:

$$\min_{\hat{\beta}} S(\hat{\beta}), \quad S(\hat{\beta}) := (Y - X\hat{\beta})'(Y - X\hat{\beta})$$

- **Step 1:** differentiate $S(\hat{\beta})$ with respect to $\hat{\beta}$:

$$\frac{\partial S(\hat{\beta})}{\partial \hat{\beta}} = -2X'Y + 2X'X\hat{\beta}$$

- **Step 2:** set partial derivative to zero and solve:

$$X'X\hat{\beta} = X'Y \Rightarrow \hat{\beta} = (X'X)^{-1}X'Y$$

-
- **Step 3:** ensure the solution $\hat{\beta}$ is in fact the minimizer by checking that the second-derivative of $S(\hat{\beta})$ is positive-definite matrix:

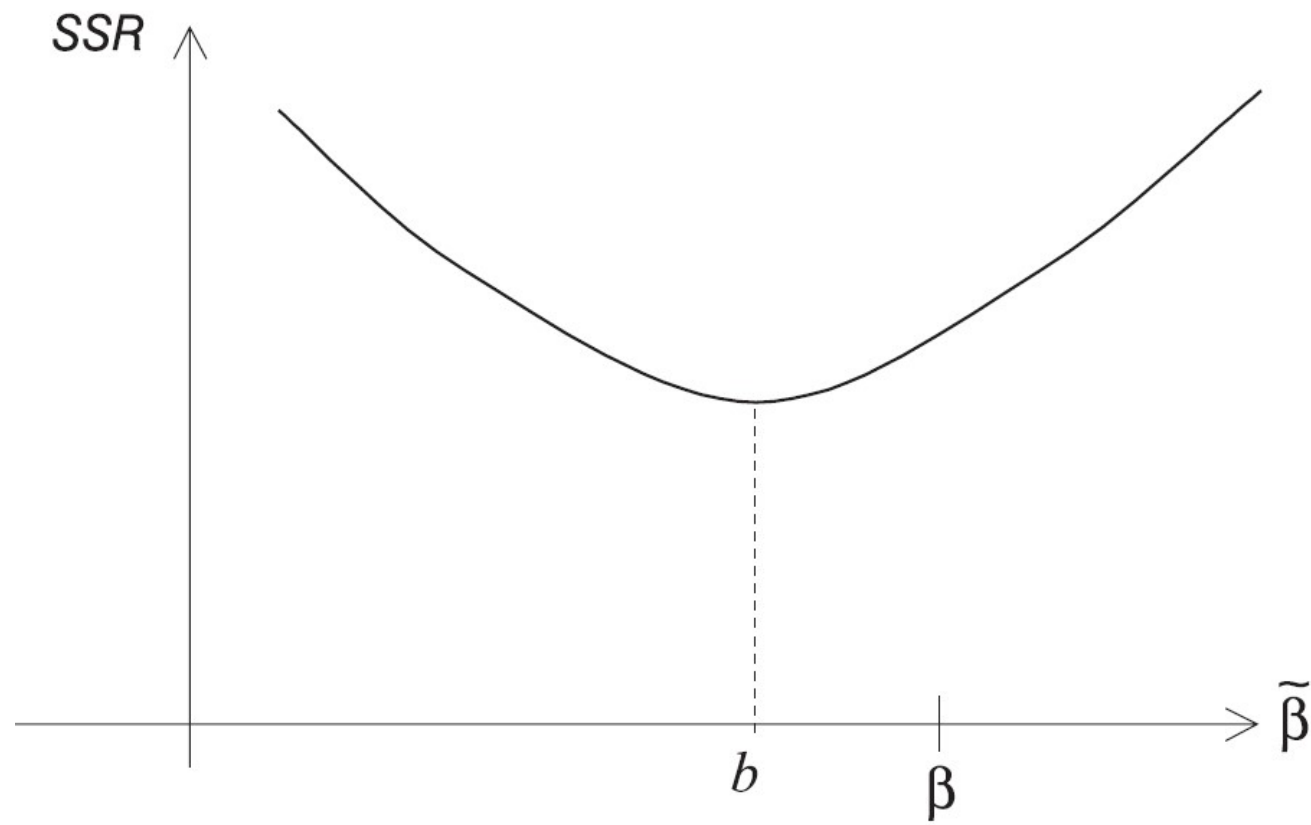
$$\frac{\partial^2 S(\hat{\beta})}{\partial^2 \hat{\beta}} = 2X'X > 0$$

- We say that a symmetric $n \times n$ square-matrix A is positive-definite if for any non-zero n -dimensional vector v , we always have

$$v'Av > 0$$

- Therefore, the OLS estimator of β for multivariate regressors is

$$\hat{\beta} = (X'X)^{-1}X'Y$$



Next: Variance Estimator

Homoskedastic Variance Estimator

- Recall in the simple linear regression case, our sample estimator for $Var(u_i)$ is given as $\frac{\sum_{i \in [n]} \hat{u}_i^2}{n-2}$
- In the multivariate case, we have a similar expression. Define $\hat{e}_i := Y_i - \hat{Y}_i$, where $\hat{Y}_i = X_i' \hat{\beta}$

- We define

$$\hat{\sigma}_\epsilon^2 := \frac{\sum_{i \in [n]} \hat{e}_i^2}{n - K - 1} = \frac{\hat{e}'\hat{e}}{n - K - 1}$$

where $K + 1$ is the dimension of β

- Under the assumption of homoskedastic variance, $\mathbb{E}[\hat{\sigma}_\epsilon^2] = Var(\epsilon_i) = \sigma^2$, an unbiased estimator
- This is not too useful for heteroskedastic data!

Heteroskedastic Variance Estimator

- The Huber-White robust variance estimator is

$$\hat{S} \equiv \widehat{Var}(\hat{\beta}) = \frac{1}{n} S_{XX}^{-1} \left(\frac{1}{n} \sum_{i \in [n]} \hat{e}_i^2 X_i X_i' \right) S_{XX}^{-1},$$

$$S_{XX} \equiv \frac{1}{n} X' X$$

- How to find $SE(\hat{\beta}_j)$?

-
- Define the vector

$$\iota_j := \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{R}^{K+1},$$

where the entry in the j th position is 1 and all other entries are 0.

- Then the sample standard error of $\hat{\beta}_j$ is

$$SE(\hat{\beta}_j) = \sqrt{\iota_j' \hat{S} \iota_j}$$

- In softwares, the sample standard-error is already computed for you.

```
. reg price industry proxy volume, robust
```

Regression with robust standard errors

Number of obs = 420
 F(2, 417) = 223.82
 Prob > F = 0.0000
 R-squared = 0.4264
 Root MSE = 14.464

	price	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
industry proxy		-1.101296	.4328472	-2.54	0.011	-1.95213	-.2504616
volume		-.6497768	.0310318	-20.94	0.000	-.710775	-.5887786
_cons		686.0322	8.728224	78.60	0.000	668.8754	703.189

- From the Stata output, $SE(\hat{\beta}_{industry_proxy}) = 0.4328472$ and $SE(\hat{\beta}_{volume}) = 0.0310318$

*Next: Inference for Individual Slope
Coefficient*

Distribution of t-statistic

- Supposed we are interested in making inference for slope parameter β_j
- We hypothesize that the true value of β_j is actually β_j^* , i.e. we want to test $H_0 : \beta_j = \beta_j^*$

- We construct our t-statistic

$$t_j := \frac{\hat{\beta}_j - \beta_j^*}{SE(\hat{\beta}_j)}$$

- t_j will be approximately $N(0, 1)$ distributed under the null hypothesis, when n is large.

Testing

- Our p -value for a two sided test equals $2\Phi(-|t_j|)$
- Reject at $1 - \alpha$ confidence-level if $p \leq \alpha$
- Example: suppose we want to test if β_j is statistically significant, i.e. $H_0 : \beta_j = 0$. We calculate $t_j = 2$. Then $p = 2\Phi(-2) = 0.0456$. For $\alpha = 0.05$ (i.e. 95% confidence-level), since $p \leq \alpha$, we reject the null and conclude that β_j is not zero, i.e. statistically significant

*Next: Joint Hypothesis Testing
under Homoskedasticity*

Motivation

- Recall that in the presense of multicollinearity, the standard errors (SE) blow up:

```
. clear
. set obs 1000
obs was 0, now 1000
. gen x=uniform()
. gen x2=x+0.001*invnorm(uniform())
. gen y=4+3*x+invnorm(uniform())
. reg y x x2
```

Source	SS	df	MS			
Model	795.758978	2	397.879489	Number of obs =	1000	
Residual	1037.3519	997	1.04047332	F(2, 997) =	382.40	
				Prob > F =	0.0000	
				R-squared =	0.4341	
				Adj R-squared =	0.4330	
Total	1833.11087	999	1.83494582	Root MSE =	1.02	

y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
x	11.21924	33.24992	0.34	0.736	-54.02861	76.46708
x2	-8.113489	33.24836	-0.24	0.807	-73.35828	57.1313
_cons	3.94279	.0645122	61.12	0.000	3.816194	4.069385

.

-
- In the above regression, 'y' depends on 'x' but not 'x2'. Since 'x' and 'x2' are highly collinear, if we add both variables and run the regression, SE blows up
 - The t-statistic for x and x2 are 0.34 and -0.24 respectively
 - If we test for slope coefficient individually using t-test, we **do not reject** $H_0 : \beta_x = 0$ and $H_0 : \beta_{x2} = 0$, concluding that both coefficients are not significantly different from zero, which we know is wrong!
 - **Remedy:** Joint testing, i.e. $H_0 : \beta_x = \beta_{x2} = 0$
 - This is also called an F-test (named after the famous statistician Ronald Fisher)
 - In fact, the p -value for the F-test gives 0.0000, so we should conclude that there is at least one slope coefficient of x or x2 that is non-trivial.

Tests involving more than one coefficient

- But what if we want to test more complicated hypotheses?
- For example, what if the null is related to more than one coefficient?
 - Example 1: $H_0 : \beta_1 = \beta_2$ vs. $H_1 : \beta_1 \neq \beta_2$
 - Example 2: $H_0 : \beta_1 + 2\beta_2 = -3$ vs. $H_1 : \beta_1 + 2\beta_2 \neq -3$
 - Example 3: $H_0 : \beta_1 = 0$ and $\beta_2 = 0$ vs. $H_1 : \beta_1 \neq 0$ or $\beta_2 \neq 0$

Caveat of t-test

- Can we just use the two t -statistics to construct two t -tests?
- Consider testing $H_0 : \beta_1 = 0$ and $\beta_2 = 0$
- Say we set the confidence level at 95%
 - Compare $\left| t_{\hat{\beta}_1} \right|$ and $\left| t_{\hat{\beta}_2} \right|$ to the 5% critical value (1.96) and reject if either one is bigger than it?
 - No!
- Why not? This test won't have the correct 'size' (or significance level)
 $\Rightarrow$ you will be wrong more than 5% of the time

Testing joint hypotheses, cont'd

- Suppose we use the two t -statistics and reject the null hypothesis if either one exceeds 1.96 in absolute value.
- Let's consider the simple case where the t -statistics are uncorrelated and therefore independent (because they are asymptotically normal)
- What, then, is the probability that you will reject the null hypothesis when the **null is in fact true**?

Testing joint hypotheses, cont'd

- We know that the null is *not* rejected only if *both*

$$\left| \hat{t}_{\beta_1} \right| \leq 1.96 \text{ and } \left| \hat{t}_{\beta_2} \right| \leq 1.96$$

- Since the t -ratios are independent here (by assumption)

$$P \left(\left| \hat{t}_{\beta_1} \right| < 1.96, \left| \hat{t}_{\beta_2} \right| < 1.96 \right) = P \left(\left| \hat{t}_{\beta_1} \right| < 1.96 \right) P \left(\left| \hat{t}_{\beta_2} \right| < 1.96 \right) = .95^2 = .9025$$

- So the probability of rejecting the null when the null is true is $\alpha = 1 - .95^2 = .0975$, not .05!
- The joint test will then have a 9.75% significance level, not 5%
- You will be wrong almost twice as often as you thought!

Testing joint hypotheses, cont'd

- If the t -statistics are not independent, we need to account for the correlation between them. In general, we will still over-reject
- So what now?
- One option would be to modify this “one at a time” method so that its size is equal to its significance level (i.e. pick a number z_α bigger than 1.96 so that $P\left(\left|t^{act}\right| < z_\alpha\right) = .975$).
- This method, called the Bonferroni test, is rarely used in practice (because it has low power) so we will not cover it in this class
- **Fortunately, we can use F test**

F-statistic

- The F -statistic exploits the fact that the t -statistics of individual coefficients are (asymptotically) normally distributed
- Suppose we want to test $H_0 : \beta_1 = 0$ and $\beta_2 = 0$
- The F -statistic combines the t -statistics $t_{\hat{\beta}_1}$ and $t_{\hat{\beta}_2}$ using the formula

$$F = \frac{1}{2} \left(\frac{t_{\hat{\beta}_1}^2 + t_{\hat{\beta}_2}^2 - 2\hat{\rho}_{t_{\hat{\beta}_1} t_{\hat{\beta}_2}} t_{\hat{\beta}_1} t_{\hat{\beta}_2}}{1 - \hat{\rho}_{t_{\hat{\beta}_1} t_{\hat{\beta}_2}}^2} \right)$$

where $\hat{\rho}_{t_{\hat{\beta}_1} t_{\hat{\beta}_2}}$ is an estimator of the correlation between the two t -statistics

- But how is this new test statistic distributed?

F-statistic, cont'd

- Consider the case where the t -statistics are uncorrelated, so that

$$F = \frac{1}{2} \left(t_{\hat{\beta}_1}^2 + t_{\hat{\beta}_2}^2 \right)$$

- Now, under H_0 , $t_{\hat{\beta}_1}$ and $t_{\hat{\beta}_2}$ are independent standard normal random variables (asymptotically) so that, F has an (asymptotic) $F_{2,\infty}$ dis.
- Recall that:
 - χ_m^2 is the sum of m squared independent standard normals
 - $F_{n,m} = \frac{\chi_n^2/n}{\chi_m^2/m}$ (where the χ^2 's are independent)
 - $F_{n,\infty} = \chi_n^2/n$ (**the average of n squared normals**)
- So (in this example) $F = \frac{1}{2} \left(t_{\hat{\beta}_1}^2 + t_{\hat{\beta}_2}^2 \right) \sim F_{2,\infty}$

F-statistic, cont'd

- A large value of F will then lead you to reject the null (since a large value of F means that $t_{\hat{\beta}_1}^2$ or $t_{\hat{\beta}_2}^2$ (or both) are large)
- In general (for this H_0), the above formula adjusts for any correlation between the t -statistics so that under the null, the F -statistic will have a $F_{2,\infty}$ distribution in large samples (regardless of the correlation in the t -statistics)
- Why is the degree of freedom of F 2 when testing $H_0 : \beta_1 = 0$ and $\beta_2 = 0$?
- The general formula for the F -statistic with q restrictions is quite complicated, so we will not present it here
- However, under the null hypothesis, the F -statistic is distributed $F_{q,\infty}$ (in large samples)

F-statistic, cont'd

- The F -statistic can be used to test both complicated hypotheses like

$$H_0 : \beta_1 = \beta_2 \text{ and } \beta_3 = 2\beta_1,$$

which has 2 restrictions, as well as simple hypotheses like

$$H_0 : \beta_1 = \beta_2 \text{ and } H_1 : \beta_1 \neq \beta_2$$

which has only 1 restriction

- Importantly, the F -statistic is automatically computed by most statistical packages with simple commands

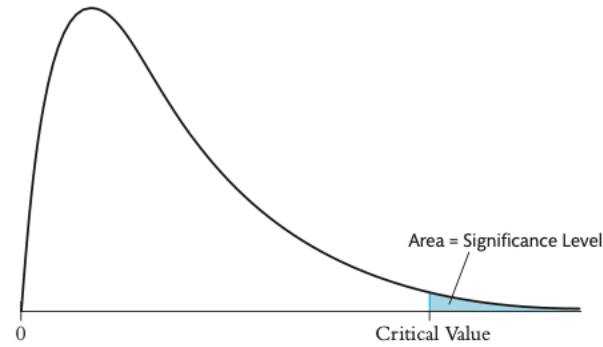
F-statistic, Caveat

- The F -statistic can be used to test any **linear** restriction or set of **linear** restrictions
- Non-linear restrictions (e.g. $H_0 : \beta_1 = \beta_2^2$) require a different methodology that we won't cover here

Using the F-statistic in practice

1. Count the number of “restrictions” under the null (degrees of freedom), call this q
2. Compute the F -statistic
3. Check the table for $F_{q,\infty}$
4. The p -value is calculated as $\Pr(F_{q,\infty} > F^{actual}) = \Pr(\chi_q^2 > q \cdot F^{actual})$

Note: Since $F_{q,\infty} = \chi_q^2/q \implies \chi_q^2 = qF_{q,\infty}$, you can also use the tables for the χ_q^2 distribution

TABLE 4 Critical Values for the $F_{m,\infty}$ Distribution

Degrees of Freedom	Significance Level		
	10%	5%	1%
1	2.71	3.84	6.63
2	2.30	3.00	4.61
3	2.08	2.60	3.78
4	1.94	2.37	3.32
5	1.85	2.21	3.02
6	1.77	2.10	2.80
7	1.72	2.01	2.64
8	1.67	1.94	2.51
9	1.63	1.88	2.41
10	1.60	1.83	2.32
11	1.57	1.79	2.25
12	1.55	1.75	2.18
13	1.52	1.72	2.13
14	1.50	1.69	2.08
15	1.49	1.67	2.04
16	1.47	1.64	2.00
17	1.46	1.62	1.97
18	1.44	1.60	1.93
19	1.43	1.59	1.90
20	1.42	1.57	1.88
21	1.41	1.56	1.85
22	1.40	1.54	1.83
23	1.39	1.53	1.81
24	1.38	1.52	1.79
25	1.38	1.51	1.77
26	1.37	1.50	1.76
27	1.36	1.49	1.74
28	1.35	1.48	1.72
29	1.35	1.47	1.71
30	1.34	1.46	1.70

This table contains the 90th, 95th, and 99th percentiles of the $F_{m,\infty}$ distribution. These serve as critical values for tests with significance levels of 10%, 5%, and 1%.

TABLE 3 Critical Values for the χ^2 Distribution

Degrees of Freedom	Significance Level		
	10%	5%	1%
1	2.71	3.84	6.63
2	4.61	5.99	9.21
3	6.25	7.81	11.34
4	7.78	9.49	13.28
5	9.24	11.07	15.09
6	10.64	12.59	16.81
7	12.02	14.07	18.48
8	13.36	15.51	20.09
9	14.68	16.92	21.67
10	15.99	18.31	23.21
11	17.28	19.68	24.72
12	18.55	21.03	26.22
13	19.81	22.36	27.69
14	21.06	23.68	29.14
15	22.31	25.00	30.58
16	23.54	26.30	32.00
17	24.77	27.59	33.41
18	25.99	28.87	34.81
19	27.20	30.14	36.19
20	28.41	31.41	37.57
21	29.62	32.67	38.93
22	30.81	33.92	40.29
23	32.01	35.17	41.64
24	33.20	36.41	42.98
25	34.38	37.65	44.31
26	35.56	38.89	45.64
27	36.74	40.11	46.96
28	37.92	41.34	48.28
29	39.09	42.56	49.59
30	40.26	43.77	50.89

This table contains the 90th, 95th, and 99th percentiles of the χ^2 distribution. These serve as critical values for tests with significance levels of 10%, 5%, and 1%.

TABLE 4 Critical Values for the $F_{m, \infty}$ Distribution

Degrees of Freedom	Significance Level		
	10%	5%	1%
1	2.71	3.84	6.63
2	2.30	3.00	4.61
3	2.08	2.60	3.78
4	1.94	2.37	3.32
5	1.85	2.21	3.02
6	1.77	2.10	2.80
7	1.72	2.01	2.64
8	1.67	1.94	2.51
9	1.63	1.88	2.41
10	1.60	1.83	2.32

TABLE 3 Critical Values for the χ^2 Distribution

Degrees of Freedom	Significance Level		
	10%	5%	1%
1	2.71	3.84	6.63
2	4.61	5.99	9.21
3	6.25	7.81	11.34
4	7.78	9.49	13.28
5	9.24	11.07	15.09
6	10.64	12.59	16.81
7	12.02	14.07	18.48
8	13.36	15.51	20.09
9	14.68	16.92	21.67
10	15.99	18.31	23.21

Question for discussion

- If you have a single test $H_0 : \beta_1 = b$ v.s. $H_1 : \beta_1 \neq b$. You can still run the F-test. In this case, $q = 1$ and

$$\text{F-stat} = t_{\hat{\beta}_1}^2.$$

What critical value will you use if the confidence level is set at 95%?

- Of course you can use the usual four methods for this single test. Will the answer you get from either of the four methods different from the one you get from the F-test?

Summary: testing multiple restrictions

- Assuming OLS Assumptions 1-4 hold (but not necessarily A5 - homoskedasticity) and a large sample size:
 - Use the F -statistic (or t -statistic if you have only one restriction)
 - When you don't have access to a computer, I will give you the critical value of the F -distribution (say, in exams)
 - If you do have access to a computer, use the proper command to compute the F -statistic and either:
 - * Use the p -value provided and reject the null if the p -value is less than the chosen significance level α .
 - * Use the tables to choose the appropriate cell.
- Now, let's look at some examples using the returns of stock

Examples

```
. reg returns volatility leverage size, robust
```

Regression with robust standard errors

Number of obs = 420
F(3, 416) = 147.20
Prob > F = 0.0000
R-squared = 0.4360
Root MSE = 14.353

returns	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]
volatility	-.2863992	.4820729	-0.59	0.553	-1.234002 .661203
leverage	.0038679	.0015807	2.45	0.015	.0007607 .0069751
size	-.6560227	.0317844	-20.64	0.000	-.7185008 -.5935446
_cons	649.5779	15.45834	42.02	0.000	619.1917 679.9641

- Let's start by testing $H_0 : \beta_1 = \beta_2$ and $\beta_3 = 2\beta_1$
- We get

```
. test (volatility=leverage) (size=2*volatility)

( 1) volatility - leverage = 0
( 2) - 2 volatility + size = 0

F( 2, 416) = 219.65
Prob > F = 0.0000
```

-
- Note that here we calculate the p -value using the $F_{2,416}$ distribution instead of the $F_{2,\infty}$ distribution discussed in Stock and Watson. My take, just use $F_{2,\infty}$.
 - The results will be essentially the same if the sample is reasonably large.

*Next: Joint Testing under
Heteroskedasticity*

Linear Hypothesis

- In general, we want to test the following:

$$H_0 : R\beta = r$$

where R is a $q \times (K + 1)$ restriction matrix, β is the K -dimensional (column) vector of interest, r is a q dimensional (column) vector.

- Note that we require $\text{Rank}(R) = q$.
- We can think of the rank of the matrix as the number of independent column vectors in the matrix.
- Recall for the previous example we wanted to test $H_0 : \beta_x = \beta_{x2} = 0$.

```

. clear
. set obs 1000
obs was 0, now 1000
. gen x=uniform()
. gen x2=x+0.001*invnorm(uniform())
. gen y=4+3*x+invnorm(uniform())
. reg y x x2

```

Source	SS	df	MS
Model	795.758978	2	397.879489
Residual	1037.3519	997	1.04047332
Total	1833.11087	999	1.83494582

Number of obs = 1000						
F(2, 997) = 382.40						
Prob > F = 0.0000						
R-squared = 0.4341						
Adj R-squared = 0.4330						
Root MSE = 1.02						

	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
y						
x	11.21924	33.24992	0.34	0.736	-54.02861	76.46708
x2	-8.113489	33.24836	-0.24	0.807	-73.35828	57.1313
_cons	3.94279	.0645122	61.12	0.000	3.816194	4.069385

```

.

```

- We can then write

$$R = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_x \\ \beta_{x2} \end{pmatrix}, \quad r = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Wald-test

- The original F-test can only handle homoskedasticity. We give a more general testing method here called the Wald-test, which is robust to heteroskedasticity.
- Under the null hypothesis, the Wald-test (denoted W) has the following property:

$$W := (R\hat{\beta} - r)' [R\hat{S}R']^{-1} (R\hat{\beta} - r) \rightsquigarrow \chi_q^2$$

where we recall

$$\hat{S} = \frac{1}{n} S_{XX}^{-1} \left(\frac{1}{n} \sum_{i \in [n]} \hat{e}_i^2 X_i X_i' \right) S_{XX}^{-1},$$
$$S_{XX} = \frac{1}{n} X'X, \quad \hat{e}_i = Y_i - X_i^\top \hat{\beta}$$

-
- How do we test?
 - Let $C_\alpha = \chi_{q,1-\alpha}^2$ be the critical-value
 - Reject the null whenever $W > C_\alpha$ at the $(1 - \alpha)$ -confidence level

Example

- Suppose we want to test $\beta_x = \beta_{x2} = 0$ as in the previous example.
- The R, β, r are given in the previous slide. We can compute $\hat{\beta}$ and $\hat{S}$ to obtain W . Let us assume our calculation gives $W = 380$.
- Then the p -value is $p = \Pr(\chi_2^2 > 380) \approx 0.0000$.
- We conclude that the event of observing $W = 380$ under the null is extremely rare. Therefore we reject the null at the 90%, 95%, 99%-confidence-level.